INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE NAME			
CLASS		INDEX NUMBER	

CHEMISTRY — 9729/01

Paper 1 Multiple Choice

15 September 2017

1 hour

Additional Materials: *Data Booklet*
Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

1 A given mass of ideal gas occupies a volume V and exerts a pressure p at 30 °C.

At which temperature will the same mass of the ideal gas occupy a volume $\frac{V}{3}$ and exert a pressure $2p$?

A 20 °C

B 20 K

C 202 °C

D 202 K

2 Which equation corresponds to the third ionisation energy of titanium (Ti)?

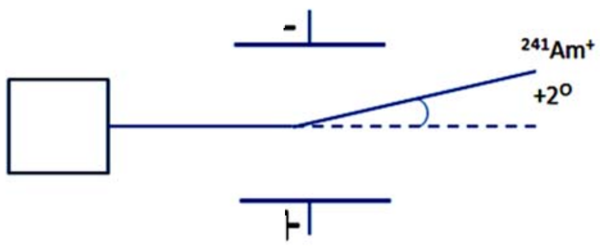
A $Ti(g) \rightarrow Ti^{3+}(g) + 3e^{-}$

B $Ti^{2+}(s) \rightarrow Ti^{3+}(g) + e^{-}$

C $Ti^{2+}(g) \rightarrow Ti^{3+}(g) + e^{-}$

D $Ti^{3+}(g) + e^{-} \rightarrow Ti^{2+}(g)$

3 A sample of the element Americium (Am) was vaporised, ionised and passed through an electric field. It was observed that a beam of $^{241}Am^{+}$ particles gave an angle of deflection of +2°.



Assuming an identical set of experimental conditions, by what angle would a beam of $^{32}S^{-}$ particles be deflected?

A +15.1°

B –15.1°

C +30.1°

D –30.1°

4 Which of the following statements describes a phenomenon which **cannot** be explained by hydrogen bonding?

A Ice floats on water.

B The boiling point of carboxylic acid increases with increasing relative molecular mass.

C 2-nitrophenol is more volatile than 4-nitrophenol.

D Ethanoic acid molecules form dimers when dissolved in benzene.

5 A tertiary amine, R_3N, reacts with boron trifluoride, BF_3 to give an addition product. Which of the following statements is **not** true?

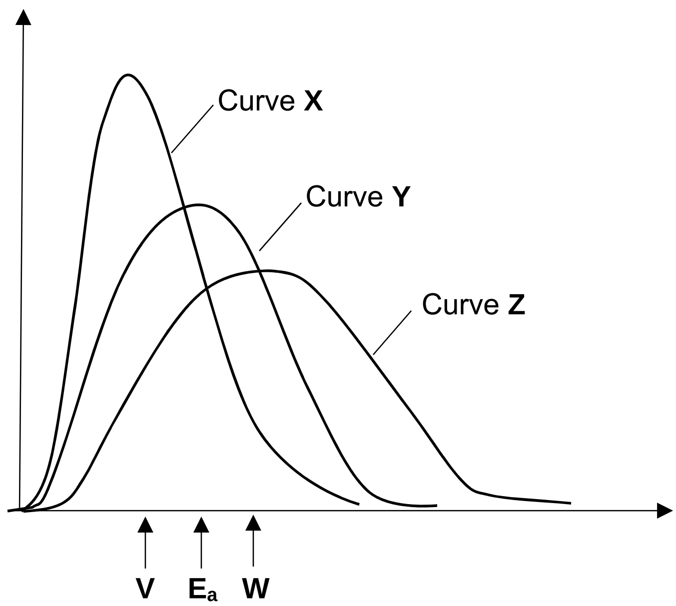
A R_3N acts as a Lewis base.

B The product is a polar molecule.

C There are six σ bonds in the product.

D The product contains a dative covalent bond.

6 The curve **Y** and the value E_a represent the distribution of energies of the molecules and the activation energy for an uncatalysed gaseous reaction.



What is a possible outcome if the reaction is catalysed?

A The distribution of energies will be given by curve **X** and the activation energy by value **V**

B The distribution of energies will be given by curve **Y** and the activation energy by value **V**.

C The distribution of energies will be given by curve **Y** and the activation energy by value **W**.

D The distribution of energies will be given by curve **Z** and the activation energy by value **W**.

7 If the rate of decay of a radioactive isotope decrease from 300 counts per minute to 37.5 counts per minute after 1 day, what is its half-life?

A 4 hours

B 6 hours

C 8 hours

D 12 hours

8 Which of the following statements does **not** describe a reaction at equilibrium?

A Forward and backward reactions occur at equal rate.

B The system must be closed.

C K_c increases as the reaction progresses.

D Concentrations of reactants and products are constant.

9 Use of the *Data Booklet* is relevant to this question.

The enthalpy change of formation of carbon monoxide and carbon dioxide are given below.

$$\Delta H_f \text{ CO} = -110 \text{ kJmol}^{-1}$$
$$\Delta H_f \text{ CO}_2 = -393 \text{ kJmol}^{-1}$$

Which of these statements are correct?

1 CO_2(g) has lower energy content than CO(g).

2 The enthalpy change of combustion of carbon is –393 kJmol^{-1}.

3 CO_2(g) is formed exothermically from CO(g).

4 A larger amount of energy is required to atomise CO_2(g) than CO(g).

A 1, 2, 3 and 4

B 1, 2 and 3 only

C 1, 2 and 4 only

D 2 and 3 only

10 A sparingly soluble calcium salt ionises in aqueous solution according to the equation given:

$$Ca_3X_2(s) \rightleftharpoons 3Ca^{2+}(aq) + 2X^{3-}(aq)$$

If the solubility product K_{sp} of Ca_3X_2 is S, what is the value of the concentration of $Ca^{2+}(aq)$ at equilibrium?

A $S^{\frac{1}{2}}$

B $\left[\dfrac{S}{108}\right]^{\frac{1}{5}}$

C $\left[\dfrac{3S}{5}\right]^{\frac{1}{2}}$

D $\left[\dfrac{9S}{4}\right]^{\frac{1}{5}}$

11 The K_{sp} of $AgCl$ and AgI are 1.80×10^{-10} $mol^2\ dm^{-6}$ and 8.3×10^{-17} $mol^2\ dm^{-6}$ respectively. Which of the following statements is correct when equal volumes of 1×10^{-4} $mol\ dm^{-3}$ aqueous $AgNO_3$ was added to a mixture containing 3.0×10^{-6} $mol\ dm^{-3}$ $BaCl_2$ and 3.0×10^{-6} $mol\ dm^{-3}$ BaI_2?

A $AgCl$ is precipitated only.

B AgI is precipitated only.

C $AgCl$ is precipitated followed by AgI.

D AgI is precipitated followed by $AgCl$.

12 The dissociation constant, K_w, for the ionisation of water, $H_2O \rightleftharpoons H^+ + OH^-$, at different temperatures is given below.

Temperature / °C	K_w / $mol^2\ dm^{-6}$
0	1.15×10^{-15}
25	1.00×10^{-14}
50	5.50×10^{-14}

What can be deduced from this information?

A Only at 25 °C are $[H^+]$ and $[OH^-]$ equal.

B The equilibrium lies furthest to the right at 0 °C.

C The forward reaction is exothermic.

D The pH of pure water decreases with temperature.

13 Calculate the standard Gibbs free energy change, $\Delta G^{\ominus}$ for the following reaction:

$$2Al\,(s) + 3Cu^{2+}(aq) \longrightarrow 2Al^{3+}(aq) + 3Cu(s)$$

A –386 kJ mol^{-1}

B –579 kJ mol^{-1}

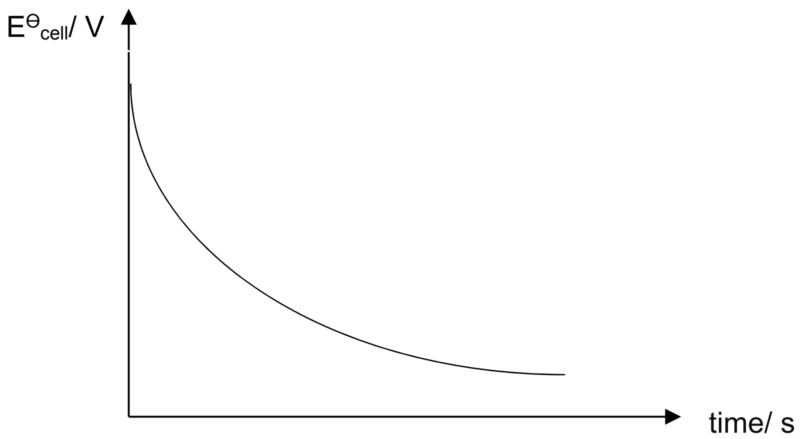
C –1045 kJ mol^{-1}

D –1158 kJ mol^{-1}

14 An experiment is carried out with the following cell.

$$Fe\,(s)\,|\,Fe^{2+}\,(aq)\,||\,Ni^{2+}\,(aq)\,|\,Ni\,(s)$$

The following graph of cell potential against time was obtained when a change was continuously made to the half-cell.



What continuous change could produce these results?

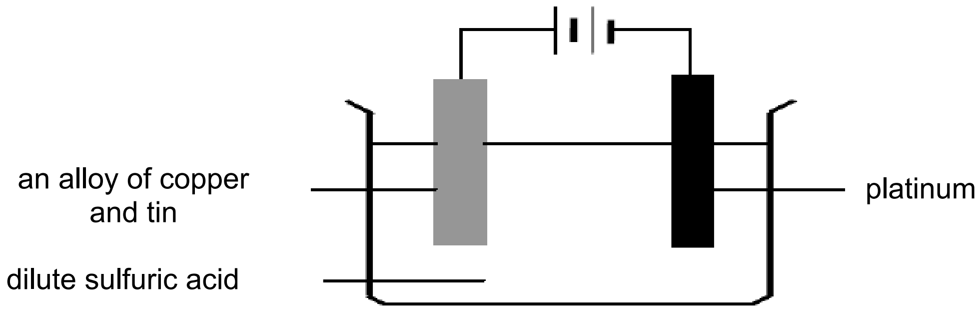
A Add nickel (II) chloride to the nickel half-cell.

B Add aqueous cyanide ions to the iron half-cell.

C Add water to the nickel half-cell.

D Increases the surface area of iron immersed in the solution.

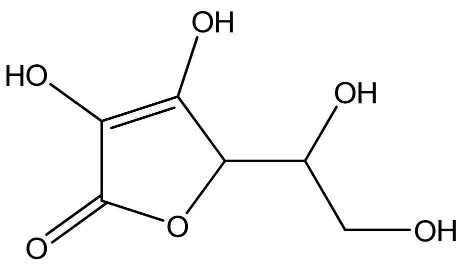
15 The circuit shown in the diagram was set up.



Which reactions will occur at the electrodes?

	anode reaction	*cathode reaction*
A	Oxygen gas is evolved.	Hydrogen gas is evolved.
B	Tin dissolves preferentially.	Hydrogen gas is evolved.
C	Copper dissolves preferentially.	Copper is deposited.
D	Copper and tin both dissolve.	Sulfur dioxide gas is evolved.

16 The diagram shows the structure of vitamin C.



vitamin C

How many stereoisomers are there in one molecule of vitamin C?

A 2 **B** 4 **C** 8 **D** 16

17 Propyne, C_3H_4, has the following structure.

$$H-C\equiv C-CH_3$$

Which row correctly describes the bonding and hybridisation in a molecule of propyne?

	number of π bonds	number of sp C atoms	number of sp^2 C atoms
A	1	1	1
B	2	2	0
C	2	2	1
D	3	3	0

18 During the nitration of benzene, a nitro group substitutes at a carbon atom. Which one of the following gives the arrangement of the bonds at this carbon atom during the reaction?

	at the start of the reaction	*in the intermediate complex*	*at the end of the reaction*
A	planar	planar	planar
B	planar	tetrahedral	tetrahedral
C	planar	tetrahedral	planar
D	tetrahedral	planar	tetrahedral

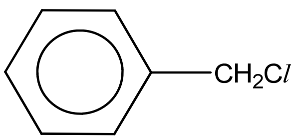
19 When a halogen compound **S** was boiled under reflux for some time with silver nitrate in a mixture of ethanol and water, little or no precipitate was seen.

Which of the following formulae could represent **S**?

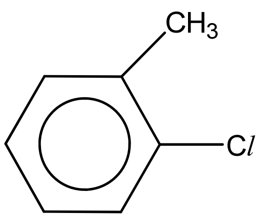
A $CH_3CH_2CHClCH_2CH_3$

B $CH_3CH_2CH_2COCl$

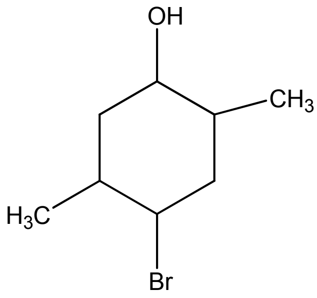
C



D



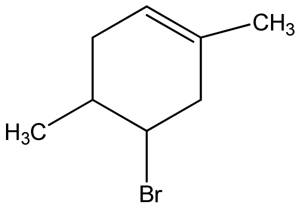
20 Compound **P** was heated with ethanolic potassium hydroxide.



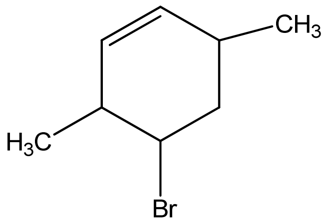
compound **P**

Which of the following would be the major product?

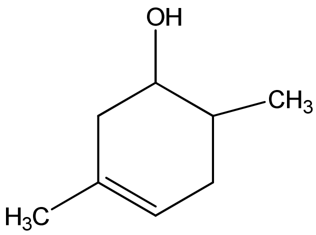
A



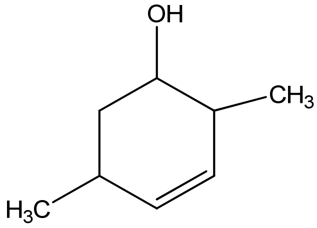
B



C



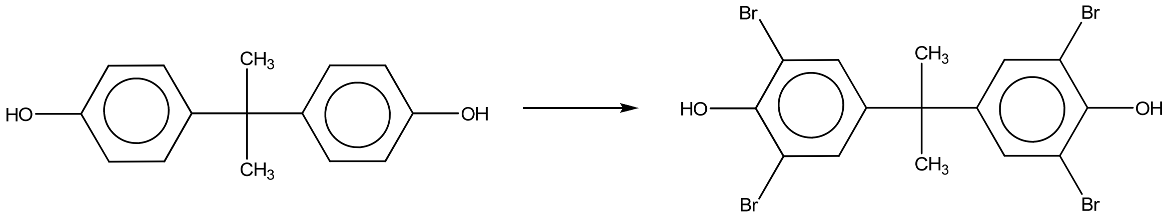
D



21 How many isomers (including both structural isomers and stereoisomers) with molecular formula $C_4H_{10}O$ liberate hydrogen gas on reaction with sodium?

A 2 **B** 3 **C** 4 **D** 5

22 Bisphenol A was used to make products such as plastic polycarbonate baby bottles and food containers. It is now regarded as toxic and has been withdrawn from use.



bisphenol A compound **Q**

Which reagent will convert bisphenol A into compound **Q**?

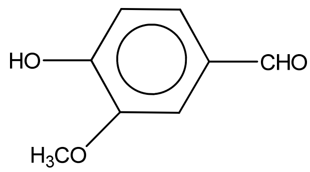
A $AlBr_3(s)$

B $Br_2(aq)$

C HBr(g)

D NaBr(aq)

23 Vanillin is the active ingredient of vanilla.



vanillin

Which of the following will be observed with vanillin?

1 Warm acidified potassium dichromate (VI) turns green.
2 2,4-dinitrophenylhydrazine reagent gives an orange precipitate.
3 A yellow precipitate is formed on warming with aqueous alkaline iodine.

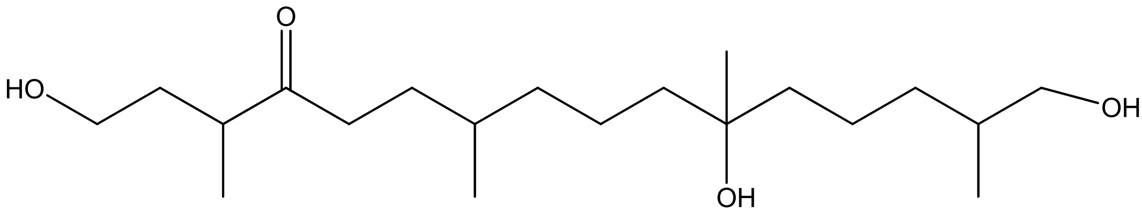
A 1 only

B 2 only

C 1 and 2 only

D 2 and 3 only

24 The mould *Phytophthora* damages many plants, destroying agricultural crops such as potatoes. A hormone-like compound called alpha1 regulates the reproduction of all species of *Phytophthora*. The structure of alpha1 is now known, giving scientist a key to the possible future eradication of the mould.



alpha1

Which of the following reagents will react with alpha1?

1 Br_2
2 $SOCl_2$
3 H_2/Pt

A 1 and 2 only

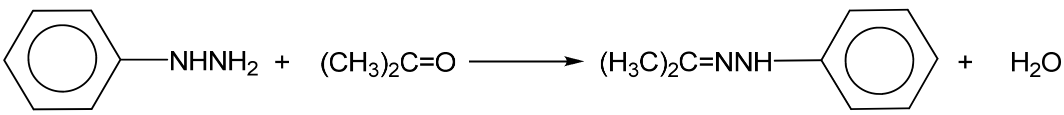
B 2 and 3 only

C 1 and 3 only

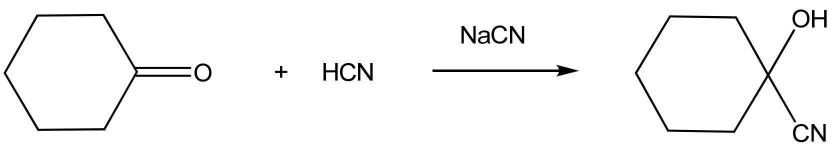
D 1, 2 and 3 only

25 Which transformations involve a nucleophile?

1



2



3 $CH_3CH_2COCl + NH_3 \longrightarrow CH_3CH_2CONH_2 + HCl$

A 1 only

B 3 only

C 1 and 2 only

D 1, 2 and 3 only

26 Which of the following properties are identical for the two enantiomers of 2-hydroxypropanoic acid, $CH_3CH(OH)COOH$?

1 $\Delta H_f^{\ominus}$

2 pK_a

3 melting point

A 2 only

B 3 only

C 1 and 2 only

D 1, 2 and 3 only

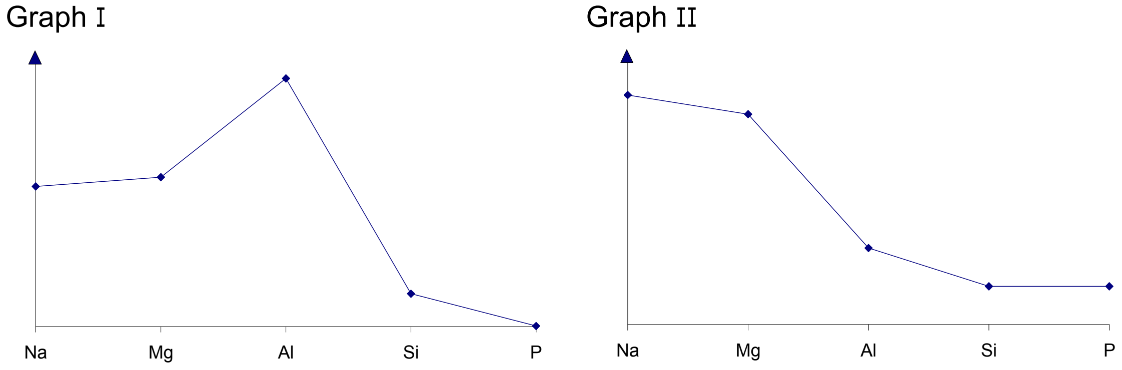
27 When organic compounds **E**, **F**, **G** and **H** are added separately to water, solutions of increasing pH values are obtained. The possible identities of compounds **E** to **H** (not necessarily in that order) are given below.

$CH_3CH_2CO_2H$ CH_3CH_2COCl $(CH_3)_2CHNH_2$ $CH_3CH_2NH_2$

Which is the correct set of identities of compounds **E**, **F**, **G** and **H**?

	E	**F**	**G**	**H**
A	$CH_3CH_2CO_2H$	CH_3CH_2COCl	$CH_3CH_2NH_2$	$(CH_3)_2CHNH_2$
B	$CH_3CH_2CO_2H$	CH_3CH_2COCl	$(CH_3)_2CHNH_2$	$CH_3CH_2NH_2$
C	CH_3CH_2COCl	$CH_3CH_2CO_2H$	$(CH_3)_2CHNH_2$	$CH_3CH_2NH_2$
D	CH_3CH_2COCl	$CH_3CH_2CO_2H$	$CH_3CH_2NH_2$	$(CH_3)_2CHNH_2$

28 The graphs below show the variation in two properties of the elements Na to P and their compounds.



Which properties are illustrated in Graphs I and II?

	Graph I	Graph II
A	electrical conductivity of the element	pH of the chloride when added to water
B	electrical conductivity of the element	pH of the oxide when added to water
C	melting point of the element	pH of the chloride when added to water
D	melting point of the element	pH of the oxide when added to water

29 Which of the following elements is expected to show the greatest tendency to form some covalent compounds?

A Barium

B Calcium

C Magnesium

D Potassium

30 Why is hydrogen iodide a stronger acid than hydrogen chloride?

A A molecule of hydrogen chloride is more polar than a molecule of hydrogen iodide.

B The enthalpy change of formation of hydrogen iodide is greater than that of hydrogen chloride.

C The covalent bond in the hydrogen iodide molecule is weaker than that in the hydrogen chloride molecule.

D The dissociation of hydrogen chloride molecules is suppressed by the stronger permanent dipole–permanent dipole interactions.

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